

Greenroom — Settlement Redesign ·

Supporting Data

Companion to `attached_assets/greenroom-settlement-redesign_1778701971579.html` .

This document does two things:

1. **Part A** — summarises everything this fork adds on top of the upstream `samay-cbh/greenroom-starter` so a reader can see what tooling produced the numbers in the report.
2. **Part B** — grounds every quantitative claim in the report against the live `/api/deal-analysis` , `/api/reports` , `/api/insights/switch-savings` and `/api/insights/switch-projected-grid` outputs. Where the live numbers strengthen, soften or outright contradict a claim, that is called out honestly — the rest of the project is built on the same standard.

All "live numbers" below were captured against the seeded `greenroom.db` snapshot at `artifacts/api-server/data/seeds/` . To re-run any of them:

```
curl -s http://localhost:5000/api/deal-analysis          | jq .
curl -s http://localhost:5000/api/reports               | jq .
curl -s "http://localhost:5000/api/insights/switch-savings?months=24" | jq .
curl -s "http://localhost:5000/api/insights/switch-projected-grid?months=24" | jq .
```

Part A — What this fork adds on top of `greenroom-starter`

A.1 The starter

Upstream had the **dataset only** — 537 settlements, 2,943 expense rows, 1,935 comp rows, 537 ticket-sales rows, 59 artists, 537 shows. No analytics tabs, no settlement wizard, no AI features, no recommendation engines. Original stack was Next.js / Vercel; this fork was migrated to a Replit pnpm monorepo (`API server` + `Greenroom` web app + `Mockup` sandbox).

The historical operational tables are **byte-for-byte unchanged** — every analytic in the app runs against the upstream 537 settlements. Demo additions (5 artists, 12 shows, 12 deals, 6 settlement history rows for "familiar" demo artists) are all upcoming shows used to demo the recommendation engines and never roll up into historical KPIs. The exclusion is enforced by every analytic gating on `shows.date ≤ today` .

A.2 New tabs (7, added in commit order)

#	TAB	WHAT IT SHOWS
1	Shows (rebuilt)	Past show grid + filters (deal type / size bucket / attention). Each row links to a Show Detail page with joined artist / agent / deal / settlement / recoups / expenses / comps.
2	Artists	Roster grid: show count, last show date, top deal type, attention badge, plus LLM-extracted positive/negative review topics.
3	Reports	KPI bundle: deal-type counts, in-app tool usage rate, settlement status distribution, dispute rate, totals (gross / payouts / recoups / comps).
4	Deal Analysis	The quantitative cross-tab: by complexity, by size bucket, by profitability, costs by category, revenue by deal type, last-24-months revenue series, full <code>dealType × sizeBucket</code> cross-tab grid with losing / disputed / attention rates per cell.
5	Needs Attention	A worklist computed from four explicit rules (<code>show_settled_no_settlement</code> , <code>notes_say_closed_but_status_open</code> , <code>disputed_recoups_but_signed</code> , <code>stale_disputed</code>).
6	Insights	Qualitative pair to Deal Analysis: same cross-tab grid, but each cell shows the dominant friction kind + LLM-clustered top-5 complaint themes. Plus the Smart Switch could have helped drill-down (per-show counterfactuals, last 3 / 6 / 12 / 24 months) and the projected grid (cross-tab recomputed under Smart Switch + Improve Deal).
7	Settings	Configures the active LLM provider / model / API key. Settings are write-only across the API (responses never echo key material). Saving invalidates the Insights cache.

A.3 New engines

- **Smart Switch** (`lib/smartSwitch.ts`) — proposes either a flat (for vs / % of net deals in \$1–5K) or a door-hybrid (\$500 floor + 60% above an expense cap, for door deals) at deal entry. Tier A/B/C/D confidence based on cell sample size + artist familiarity at the venue.
- **Smart Guaranteed Price** (`lib/smartGuarantee.ts`) — auto-generates a flat target for every upcoming non-flat deal, used for Show Detail panels and now folded into the **Improve Deal** flow.
- **Improve Deal** — caps-only structural improvements (expense cap, hospitality cap) for deals that aren't Smart-Switch-eligible. P75-flat defaults sourced from cell history.
- **Switch Savings backend** (`lib/switchSavings.ts`) — re-scores every past settled vs / % of net / door deal in a window against its Smart Switch counterfactual, returning per-show money delta + a time-saved estimate from notes/sign-off paragraph counts and disputed-recoup activity.
- **Projected Grid** — same `dealType × sizeBucket` grid as Deal Analysis, but with Smart-Switch-eligible cells re-computed under the counterfactual; muted cells (Improve-Deal territory) inherit actuals.

A.4 New pipelines + tooling

- **LLM enrichment pipeline** — per-settlement positive/negative summaries (1–2 sentences each) generated for every settlement that has notes, sign-off text, a disputed status, or a disputed recoup. Concurrency 8, idempotent. Then per-cell complaint clustering ≤ 5 themes via a second LLM pass.
- **Settle wizard** (`/shows/:id/settle`) — guided settlement entry for the simple deal shapes (flat, % of gross). The other three shapes intentionally remain "spreadsheet territory" until Phase 1+2 of the redesign land.
- **Show JSON export** (`GET /shows/:id/export`) — full per-show payload + LLM deep summary. Uses a heavier model (`claude-sonnet-4-6` / `gpt-4o`) than the default.
- **Single LLM helper** (`lib/llm.ts`) — only module that imports the Anthropic / OpenAI SDKs; everything else uses `llmGenerateText` / `llmGenerateJson` / `llmIsConfigured` . Lets the Settings tab swap providers globally with one save.
- **Calculator-style explainers** on Insights — collapsible disclosures next to both the Switch Savings section and the Projected Grid that spell out the exact formulas, eligibility, modelling assumptions, a worked example pulled from the current payload, and a footer rollup of the headline tiles.
- **Idempotent runtime DB migrations** (`db/index.ts`) — `ensureColumn` for ALTERs, `CREATE TABLE IF NOT EXISTS` for new tables; no drizzle-kit for the artifact's local libsql file.
- **Demo seed scripts** (`scripts/seedSmartSwitchDemo.ts` , `scripts/seedNewDemoProposals.ts`) — synthetic upcoming shows + familiar / first-timer artists to demo the engines.
- **Portable seed snapshot** (`data/seeds/`) — JSON exports of every fork-added row + a byte-for-byte copy of `greenroom.db` so anyone can clone and run.

A.5 New API endpoints (under `/api`)

`/shows` , `/shows/:id` , `/shows/:id/export` , `/artists` , `/reports` , `/deal-analysis` , `/needs-attention` , `/insights` , `/insights/enrich` , `/insights/switch-savings` , `/insights/switch-projected-grid` , `/insights/sgp-backtest` , `/settings/llm` (GET + POST), plus the smart-switch + improve-deal mutation routes used by Show Detail.

Part B — Where each report claim comes from (and where it diverges)

For every quantitative claim in the report we cite (a) the **live number** the API currently returns and (b) the **endpoint + cell** it comes from. Differences from the report are flagged ; matches are flagged .

B.0 Top-line scope

NUMBER	LIVE VALUE	SOURCE
Settled past shows	509	<code>/api/reports</code> → <code>totalSettlements</code>
Total deals (past)	510	<code>/api/reports</code> → <code>totalDeals</code>
Total gross box office	\$3,223,555	<code>/api/reports</code> → <code>totalGross</code>
Total payouts to artists	\$1,938,465	<code>/api/reports</code> → <code>totalToArtists</code>
Overall settlement dispute rate	4.32% (22 / 509)	<code>/api/reports</code> → <code>disputedRate</code>
In-app tool usage rate today	38.2%	<code>/api/reports</code> → <code>inAppToolUsageRate</code> (= flat + % of gross share of all deals)

B.1 Phase 1 — Door deals

Report claims: 29 door shows · 24 months · every door deal lost the venue money · 9 of 29 structurally unviable below \$2,223 gross · Pale Lake sold out 650, lost \$1,930 · hybrid would yield \$9,474 artist / \$5,983 venue.

CLAIM	LIVE VALUE	SOURCE
Door deal count	□ 29	<code>/api/deal-analysis</code> → <code>revenue.byDealType.door.count</code>
Aggregate venue net on door	□ -\$29,056 (red across the bucket)	same · <code>revenue.byDealType.door.netToVenue</code>
Door deals losing the venue money (cross-tab definition: <code>gross - payout - expenses < 0</code>)	□ 27 of 29 (93.1%) , not literally all 29	<code>/api/deal-analysis</code> → cross-tab cell <code>door \$0-1K · losingMoneyCount</code> , <code>losingMoneyRate</code>
Avg gross of the door bucket	\$4,420 (cell mean)	switch-savings basis text
Counterfactual hybrid params	□ \$500 floor + 60% above \$1,500 expense cap	<code>/api/insights/switch-savings</code> → <code>breakdown.counterfactual</code>
Pale Lake (<code>show_0207</code> , 2025-03-22) — actual artist payout	\$17,387 on \$19,296 gross / \$1,909 expenses	<code>/api/insights/switch-savings</code> <code>items[0]</code>
Pale Lake — counterfactual hybrid payout	□ \$10,020 (report says \$9,474)	same
Pale Lake — venue saving under hybrid	□ \$7,367 (report implies ≈\$5,983)	same · <code>moneySavedToVenue</code>
Pale Lake — actual venue net under door	□ \$0 by construction (door pays full pool above expenses, so <code>gross - payout - expenses ≡ 0</code>); report's -\$1,930 figure does not appear in the live data.	derived
Aggregate venue \$ saved if Smart Switch hybrid were applied to all 29 historical door deals	+\$50,220	<code>/api/insights/switch-projected-grid</code> → <code>door \$0-1K</code> cell · <code>moneySavedToVenue</code>
Loss-making nights avoided in the door bucket under hybrid	17 (27 → 10)	same cell · <code>actualLosingMoney</code> - <code>projectedLosingMoney</code>
		same cell

CLAIM	LIVE VALUE	SOURCE
Disputes avoided in the door bucket under hybrid	2 (2 → 0, by modelling assumption)	

Sub-conclusion (honest). The direction of every Phase 1 claim is supported by live data — door deals are systematically loss-making at this venue, the hybrid materially improves venue net, and the engine is already configured exactly as described (\$500 floor / 60% / \$1,500 cap, suppressed above \$15K). The exact dollar figures for Pale Lake in the report do not match the live API output and should be reconciled before publication.

B.2 Phase 2 — \$1–5K vs / % of net simplification

Report claims: 43 vs deals + 101 % of net deals · vs percentage fired 0 of 43 · 4.7% dispute rate.

CLAIM	LIVE VALUE	SOURCE
vs deals in \$1–5K	□ 137 (report cites 43; the 43 figure may correspond to a sub-window — full-window count is 137)	/api/deal-analysis → cross-tab vs \$1–5K · count
vs deals in \$0–1K	14	cross-tab vs \$0–1K · count
vs deals in \$5–15K	33	cross-tab vs \$5–15K · count
All vs deals (any size)	184	/api/reports → dealTypeCounts.vs
% of net deals total	□ 102 (report cites 101 — within rounding)	/api/reports → dealTypeCounts.percentage_of_net
Dispute rate on vs \$1–5K	□ 12.4% (report cites 4.7%; the 4.7% likely came from an earlier dataset slice)	cross-tab vs \$1–5K · disputeRate
Loss-making rate on vs \$1–5K	19.7%	same · losingMoneyRate
Aggregate venue \$ saved if Smart Switch flat were applied to all 133 modelled vs \$1–5K deals	+\$23,024 (actual payout sum \$649,911 → projected \$626,887)	/api/insights/switch-projected-grid → vs \$1–5K cell
Disputes avoided on vs \$1–5K	17 (17 → 0)	same cell
Attention items avoided across the grid	18	/api/insights/switch-projected-grid → totalAttentionAvoided

The "percentage never fired" claim — vs deals where the percentage clause out-paid the guarantee — is not currently a column on any endpoint; it needs to be derived from `actualToArtist` vs `guaranteeAmount` per settlement. The infrastructure is in place (every settled vs deal is in `/api/insights/switch-savings` 's underlying query) — adding the column is a straight loop over the same dataset.

Sub-conclusion (honest). The qualitative claim — that vs / % of net deals in the \$1–5K bucket carry materially more dispute and friction risk than equivalent flat deals — is supported by the live cross-tab (12.4% dispute rate on `vs|$1–5K` vs 3.8% on `flat|$1–5K`). The exact "0 of 43 fired" figure in the report needs to be recomputed against the current 137-deal slice before publication.

B.3 Combined headline — "If Smart Switch had been used"

Last 24 months, top 10 individual past shows where Smart Switch would have helped (`/api/insights/switch-savings?months=24`):

MONEY SAVED (TOP 10)		TIME SAVED (TOP 10)
Cumulative	\$13K+ venue-side payout reduction	5h 45m of settlement-night work avoided

Last 12 months, full counterfactual grid (`/api/insights/switch-projected-grid?months=12`) — visible inside the "If Smart Switch + Improve Deal had been used" card on `/insights` :

METRIC	LIVE VALUE	NOTES
Money saved	(depends on window — see live tile)	$\sum \text{actualPayout} - \sum \text{projectedPayout}$ over <code>switchApplies</code> cells
Loss-making nights avoided	17 (door) + flat-from-vs	per cell
Disputes avoided	19 across the grid	switched cells project to 0
Attention items avoided	18 across the grid	same modelling assumption
Modelled deals	495 of 495 candidates	every settled non-NEW-DEMO deal

B.4 Phase 3 — Insurance pricing inputs

Report claims: 2.724% dispute rate, \$384 avg contested, 257 settled shows; 99.2% of suggestion gaps $\leq$ \$400, 254 historical shows.

The live numbers we can source from the existing API:

CLAIM	LIVE VALUE	SOURCE
Settlement-level dispute rate	□ 4.32% (22 / 509) — report's 2.724% likely uses a different denominator (e.g. only deal types in scope of a flat conversion)	<code>/api/reports</code> → <code>disputedRate</code>
Total disputed-recoup value	\$7,582 across 22 disputed settlements	<code>/api/reports</code> → <code>disputedRecoupValue</code>
Implied avg contested amount per disputed settlement	\$345 ($\approx$ report's \$384 figure)	derived
Total settled (any deal type)	509	<code>/api/reports</code> → <code>totalSettlements</code>
Settlements with at least one recoup	77	<code>/api/reports</code> → <code>settlementsWithRecoups</code>

The "99.2% of suggestion gaps $\leq$ \$400" cap-sizing claim depends on backtesting the Smart Switch suggestion against the actual settled payout across the 254 in-scope historical shows. The Smart Guaranteed Price backtest at `/api/insights/sgp-backtest` is the natural home for this column; the gap distribution is already computed there per show, just not yet exposed as the " $\leq$ \$400 share" rollup.

The pricing math itself (\$21 one-sided, \$11 two-sided, \$36 Pro tier, \$10.46 expected cost, 50% target margin, \$113/month at full Crescent adoption, \$122K–\$203K platform-wide) is **internal to the report's proposal** — it is not produced by any endpoint in the current app and shouldn't be presented as a Greenroom-derived analytic.

B.5 The two report cross-checks the live app already passes cleanly

These are the ones a reviewer can verify in seconds inside the running app without any number reconciliation:

1. **Door deals are the worst-performing bucket at this venue.** Open `/deal-analysis` → cross-tab. The `door|$0-1K` cell is the only cell with a `losingMoneyRate` over 90%. No other deal type in any bucket loses money on more than half its shows.
2. **Smart Switch has the largest impact on door deals.** Open `/insights` → "If Smart Switch + Improve Deal had been used". The `door|$0-1K` cell is the single biggest contributor to total money saved (+\$50,220 on its own, vs +\$23,024 from `vs|$1-5K`). This is the live-data version of the report's central recommendation.

Recommended next steps before publishing the report

1. **Reconcile the Pale Lake numbers.** The actual show- `show_0207` figures in `/api/insights/switch-savings` should replace the \$9,474 / \$5,983 / -\$1,930 numbers in the report.
2. **Recompute the Phase-2 deal counts** (43 vs / 101 pn / 4.7% dispute) against the current dataset; the live cross-tab shows 137 vs `$1-5K` and a 12.4% dispute rate.
3. **Surface the "% never fired" column** from the existing switch-savings query so the report's strongest single claim is one click to verify.
4. **Surface the "suggestion-gap $\leq$ \$400" rollup** from the SGP backtest so the Phase-3 cap sizing has a live source.
5. **Annotate insurance-pricing figures** as proposal-side (they are modelling, not currently produced by any endpoint).

The structural argument the report makes — simplify the deal types Smart Switch already covers, then layer Improve-Deal caps and insurance on top — is fully supported by live data even before any of those reconciliations.